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Beyond Clean Success: Matched Response Profiling for Factor-Conditioned VLA Evaluation

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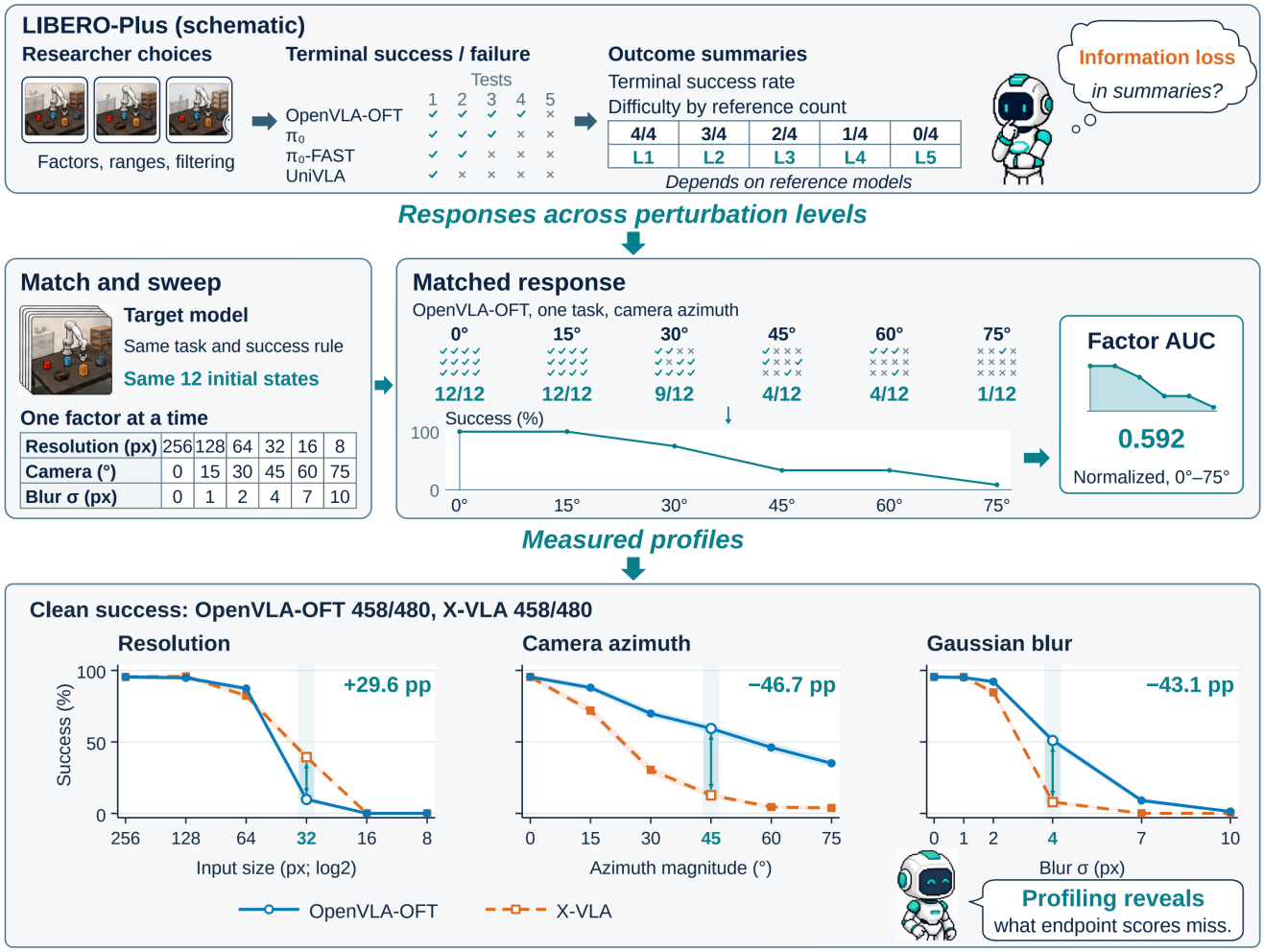


Fig. 1. Matched response profiling. Top: LIBERO-Plus reference-count construction [1]; checkmarks are illustrative. Middle: a task-specific OpenVLA-OFT camera profile with 12 matched states per level. Bottom: OpenVLA-OFT and X-VLA each achieve 458/480 clean successes but separate under visual changes. Bottom curves are suite-balanced means; ribbons are 95% matched state-block bootstrap intervals. Annotated gaps are X-VLA minus OpenVLA-OFT in percentage points.

***Abstract*— Similar aggregate benchmark scores can conceal different responses of vision-language-action (VLA) models to changing visual conditions, limiting the ability of evaluation to distinguish their performance across tasks. Task averages and endpoint success rates can hide both where models diverge and how these differences depend on the task. When observed model orderings reverse across tasks or conditions, averaging can cancel opposing differences, making equal scores an incomplete basis for comparing model behavior. We introduce *matched response profiling*, a protocol for existing benchmarks that retains task-specific success-rate curves under controlled visual changes and matched initial states, revealing response differences that aggregate scores leave unresolved. Across 46,080 rollouts from six configurations and 40 LIBERO tasks, 88.9–100% of within-task configuration pairs tied at clean and strongest-condition success exhibit different observed success rates at intermediate conditions, depending on the visual factor. In 1,440 real-robot trials with two models across three pick-and-place tasks, profiles likewise reveal observed intermediate differences hidden by endpoint ties and opposing task responses hidden by averaging. The retained information also improves configuration selection and reduces mean squared success-prediction error by 29.7–47.5% relative to reference-count predictions on held-out states of the same tasks with identical target calibration data. These findings support VLA evaluation that characterizes how performance varies across tasks and conditions, making model differences visible beyond aggregate benchmark scores.**

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I. Introduction

The same robot manipulation task can be observed from different viewpoints and at different image quality levels. Robot models combine visual observations and language instructions for planning and control [2]–[5]. Open X-Embodiment pools cross-robot experience [6], DROID broadens real-world scene coverage [7], and Octo adapts pretrained policies to new observation and action spaces [8]. Yet clean benchmark success describes only one nominal condition. Selecting a robot policy for a specific manipulation task and visual condition requires information that this nominal score alone does not provide.

Fig. 1 illustrates this gap. OpenVLA-OFT and X-VLA each achieve 458/480 matched clean successes across 40 LIBERO tasks, yet X-VLA leads by 29.6 percentage points (pp) at 32 px and trails by 46.7 pp at 45° camera azimuth. Equal nominal performance therefore hides factor-dependent response differences.

We introduce *matched response profiling*, a measurement protocol for existing benchmarks. It preserves the task and success rule, reuses the same initial states across configurations and conditions, and varies one ordered visual factor at a time. Each task–factor profile retains a success-rate curve; its normalized area under the curve (AUC) is the interval-width-weighted mean success rate over the declared factor range. Resolution, camera azimuth, and blur instantiate the protocol and let us examine what intermediate conditions reveal beyond endpoint scores.

Profiles complement nominal scores and cohort summaries, including the four-model success counts used by LIBERO-Plus [1]. These counts describe reference-cohort difficulty; profiles retain a target configuration's task-specific response along a physical coordinate. We quantify the information lost under aggregation and test its value for configuration selection and success prediction on complementary states. Aggregate and profile selection use the same factor measurements, and both predictors use identical target calibration outcomes.

The paper makes three contributions:

1) **Matched measurement.** We define a task–factor protocol that retains task and condition identity under shared initial-state blocks, enabling direct comparisons with aggregate reporting on the same outcomes. A 46,080-rollout census across six configurations and 40 tasks supports these comparisons while preserving benchmark tasks and success rules. An exhaustive state-budget sweep characterizes sensitivity to the number of evaluated initial states: using half the rollouts yields mean absolute deviations of 0.71 pp in aggregate curves and 2.97 pp in task curves from the full census.
2) **Information from intermediate conditions.** We show that intermediate conditions reveal configuration differences hidden by endpoint ties. The largest mean pairwise success-rate gap occurs at an intermediate level for each simulated factor. Across 1,440 real-robot trials on three pick-and-place tasks, two models also share endpoint rates while differing at intermediate levels. These observations support full response curves and factor-specific AUC alongside endpoint reports.
3) **Held-out validation of profile information.** We validate the retained information on held-out initial states from the same tasks. Configurations selected using task-specific profiles achieve higher mean held-out AUC than those selected using task-aggregated factor scores. With identical target calibration data, profiles reduce the mean squared error of success-probability predictions by 29.7–47.5% relative to reference-count predictions. Reference counts remain better calibrated for count-group success rates, supporting complementary reporting.

II. Related Work

a) Structured robot-policy evaluation: Robomimic supports reproducible imitation learning [9], while LIBERO evaluates transfer across four task suites [10]. ManiSkill2, RoboCasa, and RoboTwin broaden skill, household-task, and bimanual coverage [11]–[13]. VIMA separates placement, combination, object, and task generalization [14]; VLABench adds long-horizon, language-conditioned manipulation tasks to the evaluation of robot policies [15].

Factorized analysis separates sources of visual generalization error [16]; THE COLOSSEUM tests controlled environmental changes [17]. SIMPLER compares simulated and real-world policy performance [18]; RoboArena uses distributed, double-blind policy comparisons [19]; contrast sets probe language-guided task variations [20]. Agarwal et al. advocate uncertainty estimates alongside aggregate scores [21]. Our comparison fixes tasks, physical supports, and initial-state blocks while varying outcome summaries. Complementary-state tests quantify the value of retaining task and condition identity for selection and prediction.

b) Cohort-based VLA evaluation: LIBERO-Plus evaluates VLA responses to perturbations and defines L1–L5 strata from successes in a four-model construction panel [1]. These strata summarize difficulty for the reference cohort. We complement this cohort view with task-specific responses of panel-excluded configurations. Using all four-of-six reference panels in our own census, we compare count-based and profile-based predictions under identical target calibration data. The comparison separates cohort-level calibration from target-specific prediction.

III. Matched Response Profiling

A. Matched Profile Definition

A configuration combines a policy checkpoint with its execution settings. For each task, we select 12 existing LIBERO initial states [10] and reuse their initial-state–rollout-seed pairs across configurations, factors, and levels. Each task–configuration–factor profile m contains six success rates $\widehat{p}_{mk}$, each equal to successful rollouts divided by evaluated rollouts at level k. The profile retains the individual outcomes and their task, state, and condition identities. Matching the

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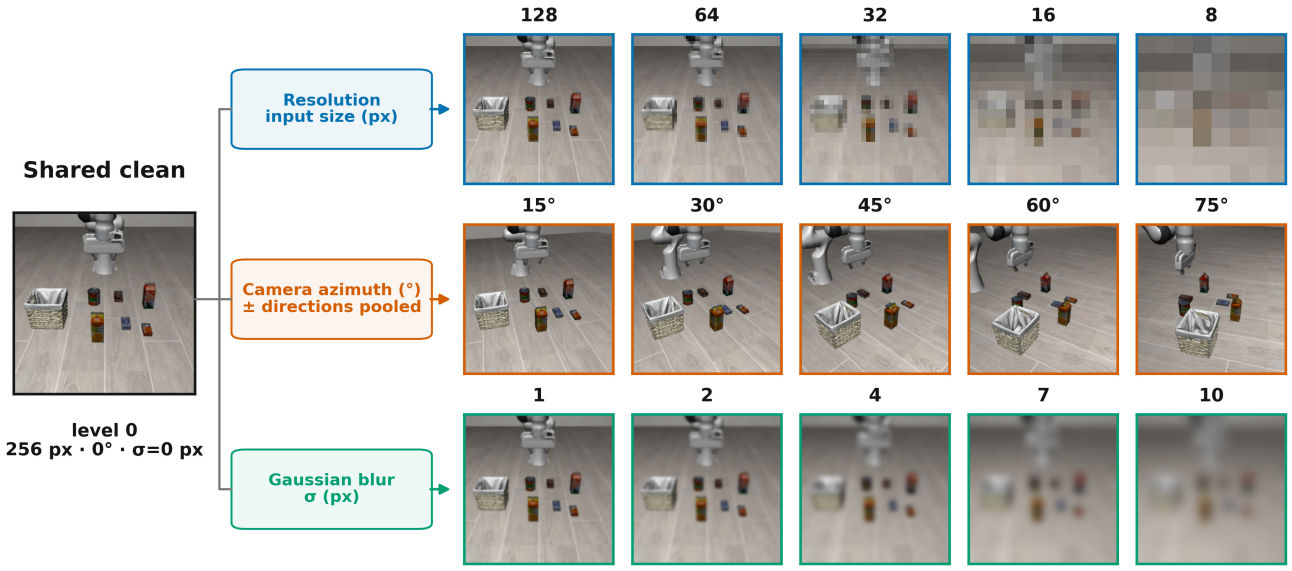


Fig. 2. Declared visual supports. One factor varies across six physical levels while the task and 12 initial states remain fixed. Camera thumbnails show one direction; measurements pool both directions. The clean condition is shared across the three factors.

state bank makes configuration and condition comparisons use the same initial conditions.

B. Ordered Physical Supports

In simulation, resolution and blur vary image detail, while camera azimuth varies viewpoint; each sweep changes one factor while preserving the task and success rule.

Resolution uses $\{256, 128, 64, 32, 16, 8\}$ pixels per side with severity $s_R(x) = \log_2(256/x)$. Camera azimuth uses magnitudes $\{0, 15, 30, 45, 60, 75\}^\circ$, which also define severity. At nonzero magnitudes, six blocks per direction are pooled; each block keeps its direction across configurations and levels. Blur uses $\sigma \in \{0, 1, 2, 4, 7, 10\}$ pixels as its severity coordinate. Resolution and blur transform both RGB streams before configuration-native preprocessing; azimuth changes the simulator agent-view camera while leaving the wrist stream unmodified. These interventions do not modify robot-state observations, language, or action interfaces. The shared clean condition gives $1+3\times5 = 16$ unique conditions. Fig. 2 shows the declared supports.

C. Success Rate and Profile AUC

Success rate measures performance at one condition; normalized AUC summarizes the same success rates over a declared factor range. For ordered severities $s_{f0}, \ldots, s_{f5}$, we integrate the piecewise-linear success-rate curve and divide by the range width:

$$A_m = \frac{1}{s_{f5} - s_{f0}} \sum_{k=0}^{4} (s_{f,k+1} - s_{fk}) \frac{\widehat{p}_{mk} + \widehat{p}_{m,k+1}}{2}. \quad (1)$$

Thus, $A_m \in [0, 1]$ is an interval-width-weighted mean success rate: a curve that remains at success rate p has AUC p. Clean success describes the nominal condition, whereas AUC includes intermediate and stronger changes. Their rankings can differ because they summarize different condition ranges. AUC uses the severity coordinates defined above, so it is not generally the arithmetic mean of six measured rates. Comparisons remain within each factor's declared range.

For example, the OpenVLA-OFT camera profile for placing alphabet soup and tomato sauce in a basket retains 72 outcomes over 12 states and six levels, with normalized AUC 0.592 (Fig. 1, middle).

D. Profile-Based Configuration Selection

We use six score blocks C to select configurations and the six complementary blocks H to evaluate them. The aggregate rule selects the highest mean AUC for each factor, averaging tasks within each LIBERO suite and the four suites equally. The profile rule selects the highest AUC separately for each task–factor pair. Both use the same score outcomes and six-level support; they differ in whether selection retains task identity. A clean rule instead selects by suite-balanced clean success on C.

For each task and factor, held-out regret is the highest configuration AUC on H minus the mean AUC of the selected top set. Tied score winners receive equal weight. The aggregate-minus-profile regret difference therefore measures the benefit of retaining task identity. Clean-versus-profile selection also changes condition coverage.

We additionally compare clean and profile selection at individual levels. Level-wise loss is the highest held-out success rate minus the mean rate of the selected top set. A rule keeps its C-selected set fixed across all levels on H.

E. Target Prediction from Counts and Profiles

For each task, block, factor, and shifted level, a four-configuration reference panel gives a count $r \in \{0, \ldots, 4\}$ of

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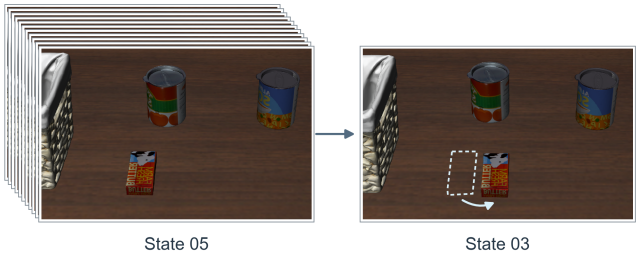


Fig. 3. Initial-state bank for placing alphabet soup and tomato sauce in a basket (LIBERO-10). State 05 tops the stack; State 03 is shown separately. The dashed outline and arrow indicate the placement change. All 12 states are reused across configurations and levels.

successful reference configurations. The target configuration is excluded from this panel.

Two predictors calibrate on identical target outcomes from C. The count predictor uses the empirical target success rate within each reference-count group, pooling tasks and shifted conditions. The profile predictor uses the target's empirical success rate for the specific task, factor, and level. On H, reference outcomes supply the count predictor's input; target outcomes serve only as evaluation labels. All five calibration count groups are nonempty in the evaluated splits.

For predicted probability $\widehat{p}_i$ and binary target outcome Y_i, the Brier score for each task–factor pair is

$$B = \frac{1}{30}\sum_{i=1}^{30}(\widehat{p}_i - Y_i)^2, \tag{2}$$

using five shifted levels and six held-out blocks. Lower values indicate more accurate outcome predictions. Count-group calibration measures the absolute gap between mean predicted and observed target success within each held-out count group.

IV. Experimental Protocol

a) Simulation configurations and task panel: We evaluate OpenVLA-OFT [22], X-VLA [23], π_0 [24], UniVLA [25], $\pi_{0.5}$ [26], and AVA-VLA [27], with reobservation/reinference intervals of 8, 30, 5, 1, 10, and 8 actions, respectively. Simulation evaluations run on NVIDIA A800 GPUs. Ten tasks from each of LIBERO-Spatial, LIBERO-Object, LIBERO-Goal, and LIBERO-10 give 40 tasks. With 12 matched states and 16 unique conditions, the six-configuration census contains 46,080 rollouts and 720 task–factor profiles; clean evaluation contributes 480 trials per configuration.

Fig. 3 illustrates placement variation within a task's matched state bank. Its clean views use the same external camera and crop, after ten settling steps and before policy execution. Episodes load the fixed initial state, execute ten zero-motion settling actions, and stop at first success or the configuration-specific horizon. Horizons for Spatial/Object/Goal/LIBERO-10 are 220/280/300/520 for OpenVLA-OFT, π_0, and AVA-VLA; 240/300/320/550 for UniVLA; 280/280/300/520 for $\pi_{0.5}$; and 800 for X-VLA in every suite. Execution settings remain fixed throughout each factor sweep.

b) Aggregation and uncertainty: Simulation curves and configuration-level AUC average states within tasks, tasks within suites, and the four suites equally. Confidence intervals and paired AUC contrasts use 10,000 matched state-block bootstrap draws [28], sharing resampled block indices across configurations, factors, and levels. Endpoint-tie fractions are descriptive.

c) Held-out tests: We evaluate all 400 balanced half-splits: C contains three blocks per camera direction, and H contains the complementary six. Each split shares block positions across configurations, tasks, factors, and levels. Estimates average splits within tasks, tasks within suites, and suites equally. For 95% intervals, 10,000 suite-stratified task-bootstrap draws resample ten tasks per suite and refit aggregate selection or count calibration in every split. Overlapping splits and reference views are not independent samples.

d) Reference panels: The analysis uses 7,200 shifted items: 40 tasks, three factors, five levels, and 12 blocks. All 15 four-of-six reference panels and their two excluded targets give 30 overlapping views. Brier scores average views within tasks. Prediction bands show the interquartile range of the 40 task means after averaging splits and views, not confidence intervals. Count-group absolute errors average counts, splits, and views equally. The census excludes π_0-FAST, so the comparison evaluates a four-model count summary rather than the original LIBERO-Plus panel.

e) Rollout-budget sensitivity: We evaluate 2, 4, 6, 8, 10, or 12 states per task, balanced across camera directions, while retaining all configurations, tasks, and conditions. These budgets enumerate 36, 225, 400, 225, 36, and 1 subsets, respectively; subset positions are shared across all comparisons. Rollout cost counts the shared clean condition once. Absolute curve-point differences from the 12-state census are computed after task averaging for aggregate curves and before averaging for task curves. We also compare the 15 configuration-pair AUC orders per factor and the best-AUC configuration set for each factor, retaining ties. Metrics average comparisons within each subset and then subsets. The full reference contains the subset outcomes, so this retrospective analysis measures sensitivity within the existing state bank.

V. Results

A. Profiles Reveal Condition-Dependent Responses

OpenVLA-OFT and X-VLA both achieve 458/480 clean successes, yet their factor-specific responses differ (Fig. 1 and Table I). The X-VLA-minus-OpenVLA-OFT AUC difference is $+0.0513$ for resolution, -0.3188 for camera azimuth, and -0.1474 for blur. Matched state-block 95% intervals are $[0.0408, 0.0617]$, $[-0.3379, -0.2992]$, and $[-0.1582, -0.1366]$. X-VLA has higher AUC for resolution, whereas OpenVLA-OFT has higher AUC for camera azimuth and blur. These orderings show why equal nominal scores do not resolve configuration choice under visual changes.

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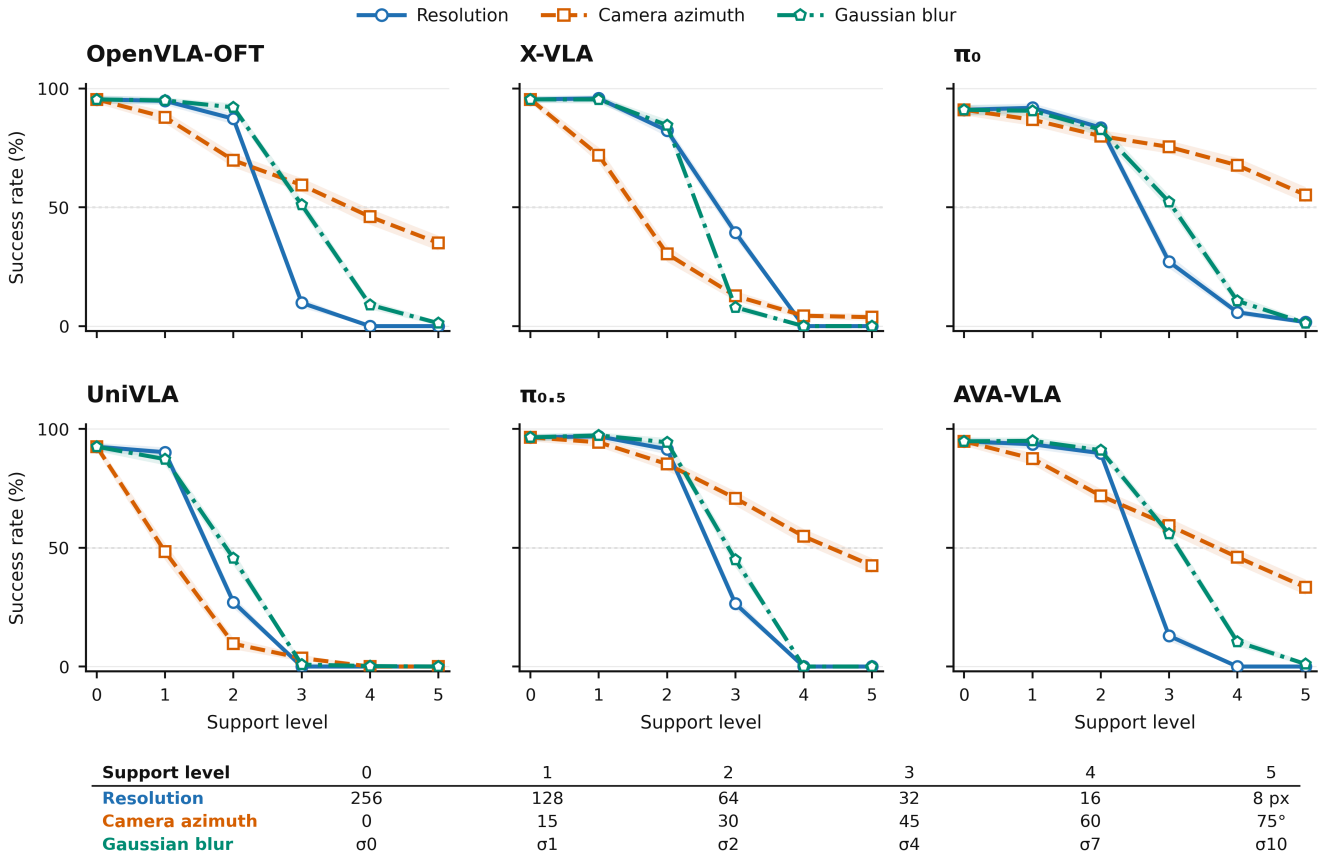


Fig. 4. Response atlas for six configurations on 40 LIBERO tasks, with 12 matched states per task. Curves are suite-balanced means; ribbons are 95% matched state-block bootstrap intervals. Colors, markers, and line styles identify factors. The bottom key maps support indices to physical values; equal indices do not imply equal perturbation strength across factors.

TABLE I

CLEAN SUCCESSES/TRIALS AND NORMALIZED FACTOR-SPECIFIC AUC FOR SIX CONFIGURATIONS ON 40 TASKS WITH 12 MATCHED STATES EACH; HIGHER IS BETTER.

Configuration	Clean	Resolution	Camera	Blur
OpenVLA-OFT	458/480	0.4792	0.6567	0.4372
X-VLA	458/480	**0.5304**	0.3379	0.2898
π_0	437/480	0.5094	**0.7663**	0.4247
UniVLA	444/480	0.3271	0.2154	0.2047
$\pi_{0.5}$	**463/480**	0.5260	0.7494	0.3996
AVA-VLA	455/480	0.4873	0.6577	**0.4514**

Across the six configurations, the configuration with the highest AUC differs by visual factor (Fig. 4 and Table I). X-VLA has the highest AUC for resolution, π_0 for camera azimuth, and AVA-VLA for blur. These rankings describe performance over each factor's declared range, supporting factor-specific reporting alongside clean success.

Intermediate measurements reveal task-specific response differences beyond clean and strongest-condition success rates (Fig. 5). For each factor, six configurations form $\binom{6}{2} = 15$ unordered pairs; comparing each pair on all 40 tasks gives 600 configuration-pair–task comparisons. We retain comparisons with equal empirical success rates at clean and the strongest setting (8 px, 75°, or blur $\sigma = 10$ px). The factor-specific endpoints yield 359 resolution, 75 camera-azimuth, and 335 blur comparisons. Each success rate is measured over the same 12 initial states for its task. A difference can be as small as one success out of 12; comparisons share tasks and configurations.

Among these endpoint ties, 319 (88.9%), 75 (100%), and 323 (96.4%), respectively, differ at one or more intermediate levels (top); red segments mark pairs with identical rates at all levels. The bottom panel uses the same factor-specific denominators to show the fraction that differs at each level. These fractions peak at 62.1% for resolution at 32 px, 77.3% for camera azimuth at 30°, and 74.3% for blur at $\sigma = 4$. A comparison can differ at several levels, while the top panel counts it once. These intermediate measurements locate response gaps that endpoint scores miss.

At the aggregate level, intermediate conditions also reveal the largest mean success-rate gaps between configurations. We average success rates over the 40 tasks for each configuration and then average absolute gaps over all 15 configuration pairs. The mean gap peaks at 23.21 pp at 64 px, 35.58 pp at 45°, and 27.61 pp at blur $\sigma = 4$. Endpoint

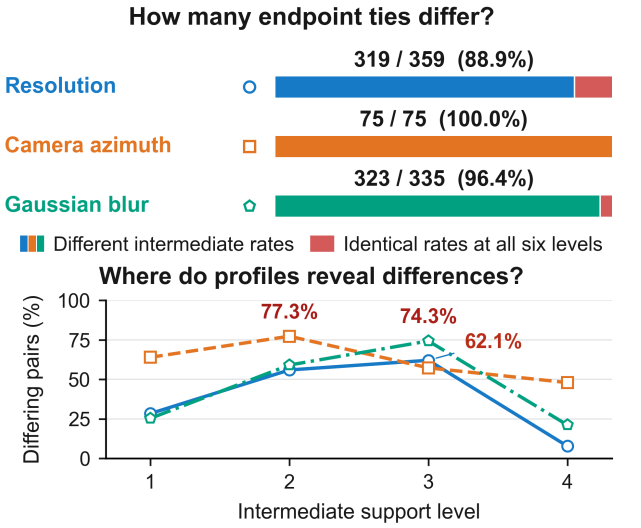


Fig. 5. Intermediate differences among pairs tied at clean and strongest conditions: 40 tasks, 12 states each. Top: fraction differing at any intermediate level; bottom: at each level. Denominators: 359 resolution, 75 camera, and 335 blur comparisons.

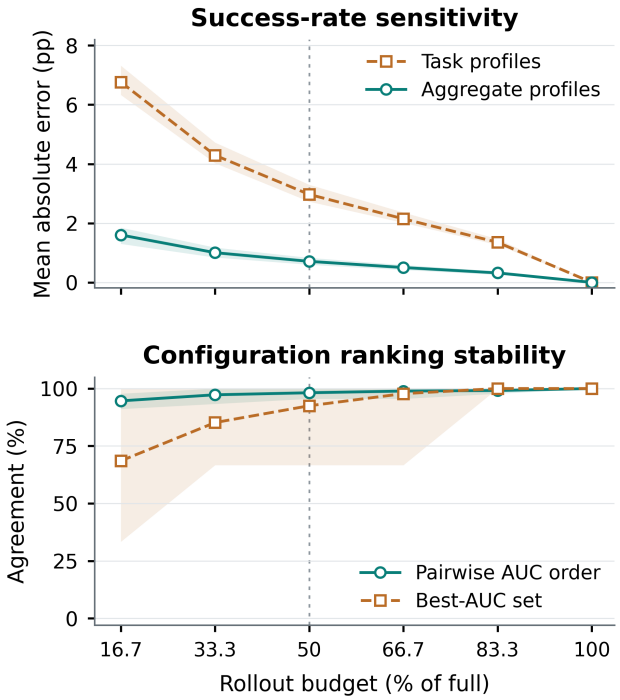


Fig. 6. Budget sensitivity on six configurations and 40 tasks relative to the 12-state census. Top: mean absolute curve errors; bottom: pairwise AUC order and factor-level best-set agreement. Lines average direction-balanced subsets; bands show their 5th–95th percentiles, not confidence intervals.

reports therefore omit the conditions with the largest mean configuration gaps on this panel.

At the task level, we exclude configuration-pair comparisons tied in either clean success or factor AUC. The two metrics reverse the ordering in 75/235 remaining resolution comparisons (31.9%), 85/235 camera comparisons (36.2%), and 90/236 blur comparisons (38.1%).

a) Sensitivity to rollout budget: Task curves are more sensitive to state count than aggregate curves (Fig. 6). At half budget (23,040 rollouts, vertical line), mean absolute point discrepancies are 0.71 pp after task averaging and 2.97 pp before averaging. Individual task-point discrepancies across subsets at this budget have a 95th percentile of 16.67 pp and reach 50 pp. Full-budget errors are zero because the census is the reference.

Half-budget agreement averages 98.16% for aggregate pairwise AUC orders and 92.50% for factor-level best-AUC sets. Resolution best-set agreement is 84.75%; changes chiefly involve X-VLA and $\pi_{0.5}$, separated by 0.44 pp in full-census AUC. All 400 subsets preserve the OpenVLA-OFT/X-VLA ordering within each factor. Broad aggregate differences persist with fewer states, while task rates and close rankings are more sensitive to sampling.

B. Profiles Improve Held-Out Selection and Prediction

a) Configuration selection: Retaining task identity lowers held-out selection regret (Table II). Agg. and Profile use identical six-state score data and six-level AUCs, selecting factor-wide and task-specific top sets, respectively. Clean changes condition coverage by using aggregate clean success. All select among six configurations and evaluate six complementary states of the same tasks. Regret is reported on a 0–100 scale; ΔR is Agg. minus Profile in pp.

Profile reduces mean regret relative to Agg. by 6.87, 5.34, and 4.58 pp for resolution, camera azimuth, and blur.

TABLE II

HELD-OUT AUC REGRET (%, LOWER IS BETTER) ON 40 TASKS. ΔR: AGG. MINUS PROFILE (PP). BRACKETS GIVE 95% TASK-BOOTSTRAP INTERVALS.

Factor	Clean	Agg.	Profile	ΔR [95% CI]	Tasks $\Delta R > 0$
Resolution	9.74	8.26	1.39	6.87 [4.12, 8.28]	33/40
Camera	13.75	7.52	2.18	5.34 [2.41, 8.22]	27/40
Blur	11.84	7.02	2.44	4.58 [1.71, 7.44]	23/40

Paired 95% suite-stratified task-bootstrap intervals are above zero, with positive task-mean reductions in 33/40, 27/40, and 23/40 tasks. Relative to Clean, Profile lowers level-wise success-rate loss at 17 of 18 conditions; at blur $\sigma = 1$, loss rises from 2.14 to 2.71 pp. Retaining task identity improves choices under the same factor measurements, with gains that vary across tasks and conditions.

b) Target-specific prediction: Profiles lower held-out prediction error by retaining task and condition identity (Fig. 7). The count predictor pools outcomes within reference-count groups; profiles retain task–factor–level rates. Both calibrate on identical target outcomes from six states C and predict six complementary states H of the same tasks. Held-out reference outcomes supply the count covariate; held-out target outcomes are evaluation labels.

Profiles lower mean Brier error at all 15 shifted factor–level combinations. Averaging each predictor's error equally over the five levels gives relative reductions of 47.5%

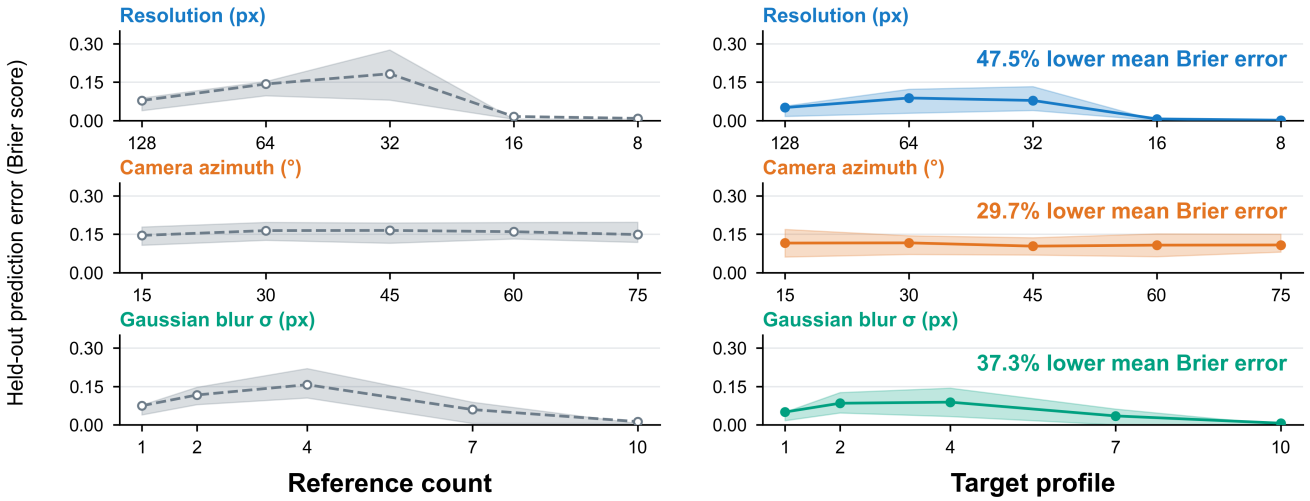


Fig. 7. Held-out prediction on 40 tasks: reference-count estimates (left) versus task–factor–level profiles (right). Both use identical six-state target calibration data and predict six complementary states. Lines show mean Brier error (lower is better); bands show task IQRs after averaging splits and reference views, not confidence intervals. Percentages give relative reductions in factor-average Brier error.

for resolution, 29.7% for camera azimuth, and 37.3% for blur. Paired absolute reductions are 0.0407 $[0.0316, 0.0493]$, 0.0466 $[0.0366, 0.0546]$, and 0.0313 $[0.0234, 0.0384]$, respectively; brackets give 95% suite-stratified task-bootstrap intervals. Task-average error decreases in 38/40, 39/40, and 38/40 tasks under the same target calibration data.

Task dispersion changes unevenly: seven level-wise IQRs narrow, three are unchanged, and five widen. Reference counts retain an ordered target-success trend in all 30 panel–target views and yield lower count-group absolute error (1.45 versus 2.13 pp for profiles). This metric evaluates average success within count groups, whereas Brier error evaluates individual predictions. A group mean can remain calibrated while hiding task-dependent responses. Profiles better predict individual outcomes; reference counts better estimate average success within count groups.

C. Real-Robot Response Profiling

Fig. 8 evaluates OpenVLA-OFT and $\pi_{0.5}$ placing an apple, banana, or bell pepper in a pan. Both policies were trained on NVIDIA A800 GPUs using the same 50 demonstration episodes per task, with randomized object placements. Real-robot evaluations use an AgileX Cobot Magic 1.0 platform, an Intel RealSense D435 camera, 30 Hz actions, and an NVIDIA GeForce RTX 4090 GPU. Each task–model–factor–level condition has 20 trials with placements randomized among three locations. Success is assessed manually. Translucent silhouettes and dashed ellipses illustrate

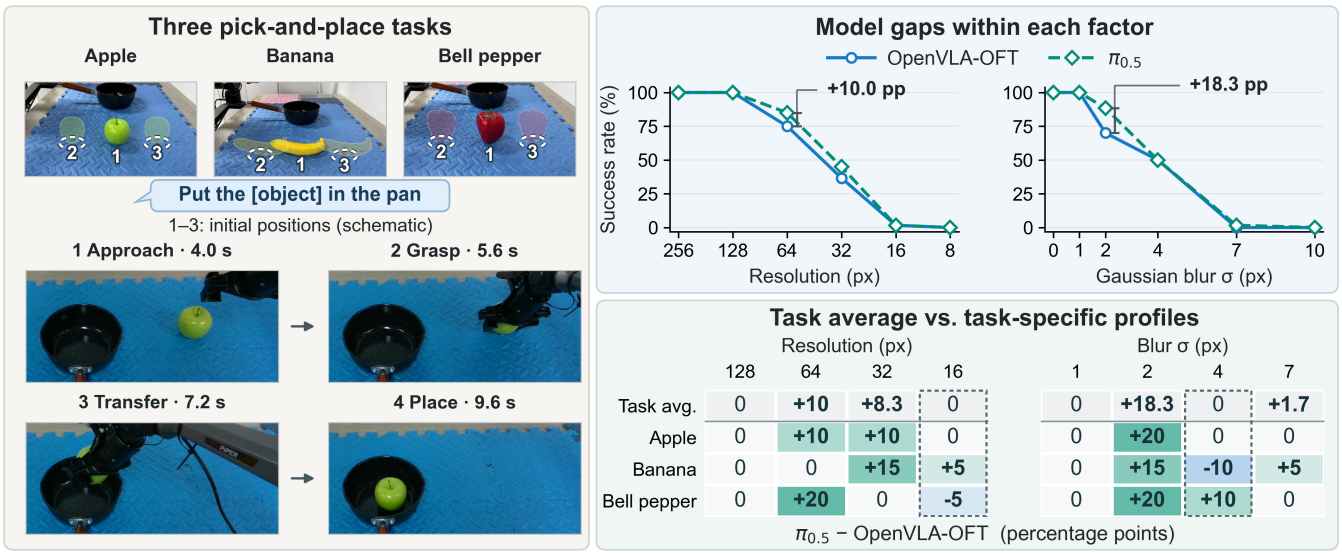


Fig. 8. Real-robot profiling on three pick-and-place tasks. Left: initial placements and one apple rollout. Right: success curves (60 trials per condition) and task-specific gaps (20 trials per task and model). Gaps are $\pi_{0.5}$ minus OpenVLA-OFT (pp); gray rows average tasks. Dashed boxes mark opposing gaps with zero mean.

placement alternatives. Resolution and blur use the supports in Fig. 2; separate clean trials per factor give 1,440 trials. Aggregate curves pool 60 trials per condition. Frame labels show video time.

Both models achieve 20/20 clean successes and 0/20 at the strongest condition in every task–factor comparison. All six comparisons nevertheless differ at one or more intermediate levels. At 64 px, pooled success is 85.0% for $\pi_{0.5}$ and 75.0% for OpenVLA-OFT; at blur $\sigma = 2$ px, it is 88.3% and 70.0%, respectively. These are the largest pooled model gaps within each factor: 10.0 and 18.3 pp. Full-support AUC is 0.5633 versus 0.5267 for resolution and 0.4125 versus 0.3800 for blur, using Eq. (1).

Task averaging can erase opposing responses even at a measured intermediate condition. At blur $\sigma = 4$ px, both models pool to 50.0% success, yet OpenVLA-OFT leads by 10 pp on banana and trails by 10 pp on bell pepper. The lower-right panel of Fig. 8 directly compares the task-averaged gaps (gray rows) with all 24 task-specific intermediate gaps. The 16 px condition also averages to zero despite opposing 5 pp gaps on banana and bell pepper, each corresponding to one success. The zero pooled gap masks observed differences of opposite signs across tasks.

VI. Discussion and Scope

At half budget, aggregate curves show a mean absolute deviation of 0.71 pp from the full census. Task rates and close rankings are more sensitive to state count, so budgets should match the intended comparison.

Counts summarize cohort difficulty; profiles retain task-specific responses. Both predictors use identical target calibration outcomes; the count predictor additionally requires reference outcomes on held-out states. This comparison fixes target calibration, not total rollout cost.

The evidence covers six configurations and 40 LIBERO tasks under single-factor changes, plus two models on three real-robot tasks. Simulation bootstrap intervals describe state or task variation within this fixed panel; real-robot rates use randomized placements. AUC depends on the declared factor range, and count-based predictions on the reference cohort. New benchmarks need controlled changes and matched resets; unseen tasks and coupled perturbations require additional measurements.

VII. Conclusion

Matched response profiling uses matched initial states to expose task-specific differences hidden by equal observed clean and strongest-condition success rates. Evidence spans six configurations on 40 LIBERO tasks and 1,440 trials on three real-robot tasks. On complementary states of measured tasks, profiles lower selection regret and reduce mean Brier error by 29.7–47.5% with identical target calibration data. Budget sweeps quantify curve and ranking sensitivity to state count. These findings support reporting task-specific profiles alongside aggregate scores.

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